

Overview of materials for High Density Polyethylene (HDPE), Extruded

Categories: [Polymer](#); [Thermoplastic](#); [Polyethylene \(PE\)](#); [High Density \(HDPE\)](#); [High Density Polyethylene \(HDPE\), Extruded](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "High Density Polyethylene (HDPE), Extruded". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

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Physical Properties	Metric	English	Comments
Density	0.933 - 1.27 g/cc	0.0337 - 0.0459 lb/in ³	Average value: 0.958 g/cc Grade Count:182
Water Absorption	0.0100 - 0.300 %	0.0100 - 0.300 %	Average value: 0.113 % Grade Count:7
Viscosity Test	300 - 500 cm ³ /g	3.00 - 5.00 dl/g	Average value: 409 cm ³ /g Grade Count:7
Environmental Stress Crack Resistance	10.0 - 10000 hour	10.0 - 10000 hour	Average value: 1000 hour Grade Count:58
	200 - 200 hour @Temperature 50.0 - 50.0 °C	200 - 200 hour @Temperature 122 - 122 °F	Average value: 963 hour Grade Count:7
	139 - 139 hour @Temperature 50.0 - 50.0 °C	139 - 139 hour @Temperature 122 - 122 °F	Average value: 963 hour Grade Count:3
	139 - 139 hour @Thickness 2.00 - 2.00 mm	139 - 139 hour @Thickness 0.0787 - 0.0787 in	Average value: 963 hour Grade Count:3
Oxidative Induction Time (OIT)	10.0 - 200 min	10.0 - 200 min	Average value: 62.5 min Grade Count:4
	20.0 - 120 min @Temperature 200 - 210 °C	20.0 - 120 min @Temperature 392 - 410 °F	Average value: 51.0 min Grade Count:10
Linear Mold Shrinkage	0.00500 - 0.0400 cm/cm	0.00500 - 0.0400 in/in	Average value: 0.0137 cm/cm Grade Count:6
Melt Flow	0.0400 - 63.0 g/10 min	0.0400 - 63.0 g/10 min	Average value: 4.28 g/10 min Grade Count:181
Ash	0.0400 - 0.0600 %	0.0400 - 0.0600 %	Average value: 0.0553 % Grade Count:14
	<= 2.00 % @Temperature 600 - 600 °C	<= 2.00 % @Temperature 1110 - 1110 °F	Average value: 2.00 % Grade Count:1

Mechanical Properties	Metric	English	Comments
Hardness, Rockwell R	80.0 - 112	80.0 - 112	Average value: 91.3 Grade Count:4
Hardness, Shore D	55.0 - 69.0	55.0 - 69.0	Average value: 62.2 Grade Count:72
Ball Indentation Hardness	40.0 - 58.0 MPa	5800 - 8410 psi	Average value: 46.8 MPa Grade Count:11
Tensile Strength, Ultimate	15.2 - 45.0 MPa	2200 - 6530 psi	Average value: 27.2 MPa Grade Count:68
	11.0 - 25.0 MPa @Temperature -30.0 - 70.0 °C	1600 - 3630 psi @Temperature -22.0 - 158 °F	Average value: 20.3 MPa Grade Count:1
Tensile Strength, Yield	2.69 - 60.7 MPa	390 - 8800 psi	Average value: 24.5 MPa Grade Count:118

	6.89476 - 30.0 MPa @Temperature -30.0 - 70.0 °C	1000.00 - 4350 psi @Temperature -22.0 - 158 °F	Average value: 20.6 MPa Grade Count:1
Film Elongation at Break, MD	400 - 600 %	400 - 600 %	Average value: 500 % Grade Count:4
Film Elongation at Break, TD	450 - 650 %	450 - 650 %	Average value: 550 % Grade Count:4
Elongation at Break	3.00 - 1900 %	3.00 - 1900 %	Average value: 610 % Grade Count:113
	200 - 500 % @Temperature -30.0 - 70.0 °C	200 - 500 % @Temperature -22.0 - 158 °F	Average value: 367 % Grade Count:1
Elongation at Yield	5.00 - 14.0 %	5.00 - 14.0 %	Average value: 9.08 % Grade Count:35
	4.00 - 9.00 % @Temperature -30.0 - 70.0 °C	4.00 - 9.00 % @Temperature -22.0 - 158 °F	Average value: 5.67 % Grade Count:1
Modulus of Elasticity	0.483 - 1.75 GPa	70.0 - 254 ksi	Average value: 1.03 GPa Grade Count:35
Flexural Yield Strength	16.5 - 91.0 MPa	2400 - 13200 psi	Average value: 46.6 MPa Grade Count:9
Flexural Modulus	0.500 - 4.83 GPa	72.5 - 700 ksi	Average value: 1.20 GPa Grade Count:81
Secant Modulus	0.531 - 1.24 GPa	77.0 - 180 ksi	Average value: 0.788 GPa Grade Count:4
Izod Impact, Notched	0.200 - 7.50 J/cm	0.375 - 14.1 ft-lb/in	Average value: 2.30 J/cm Grade Count:31
	0.700 - 1.10 J/cm @Temperature -40.0 - -40.0 °C	1.31 - 2.06 ft-lb/in @Temperature -40.0 - -40.0 °F	Average value: 0.900 J/cm Grade Count:2
Izod Impact, Unnotched	1.00 - 5340 J/cm	1.87 - 10000 ft-lb/in	Average value: 2.24 J/cm Grade Count:6
Izod Impact, Notched (ISO)	3.00 - 3.00 kJ/m² @Temperature -20.0 - -20.0 °C	1.43 - 1.43 ft-lb/in² @Temperature -4.00 - -4.00 °F	Average value: 3.00 kJ/m² Grade Count:1
Charpy Impact, Notched	0.400 - 10.5 J/cm²	1.90 - 50.0 ft-lb/in²	Average value: 4.30 J/cm² Grade Count:4
	0.220 - 0.800 J/cm² @Temperature -30.0 - -30.0 °C	1.05 - 3.81 ft-lb/in² @Temperature -22.0 - -22.0 °F	Average value: 0.518 J/cm² Grade Count:4
Tensile Impact Strength	70.0 - 480 kJ/m²	33.3 - 228 ft-lb/in²	Average value: 224 kJ/m² Grade Count:9
	90.0 - 280 kJ/m² @Temperature -30.0 - -30.0 °C	42.8 - 133 ft-lb/in² @Temperature -22.0 - -22.0 °F	Average value: 178 kJ/m² Grade Count:12
Tensile Impact	68.3 - 152 J/cm	128 - 285 ft-lb/in	Average value: 90.2 J/cm Grade Count:7
Coefficient of Friction	0.200 - 0.280	0.200 - 0.280	Average value: 0.240 Grade Count:4
Tear Strength Test	17.0 - 29.4	17.0 - 29.4	Average value: 21.6 Grade Count:11
Elmendorf Tear Strength MD	20.4 - 33.6 g	20.4 - 33.6 g	Average value: 23.9 g Grade Count:4
Elmendorf Tear Strength TD	40.8 - 204 g	40.8 - 204 g	Average value: 99.4 g Grade Count:4
Film Tensile Strength at Break, MD	45.0 - 55.0 MPa	6530 - 7980 psi	Average value: 47.8 MPa Grade Count:4
Film Tensile Strength at Break, TD	40.0 - 46.0 MPa	5800 - 6670 psi	Average value: 42.3 MPa Grade Count:4

Electrical Properties	Metric	English	Comments
Electrical Resistivity	1.00 - 1.00e+20 ohm-cm	1.00 - 1.00e+20 ohm-cm	Average value: 4.17e+18 ohm-cm Grade Count:25
Surface Resistance	100 - 1.00e+17 ohm	100 - 1.00e+17 ohm	Average value: 2.00e+16 ohm Grade Count:6

Dielectric Constant	2.00 - 2.60	2.00 - 2.60	Average value: 2.33 Grade Count:17
Dielectric Strength	11.8 - 150 kV/mm	300 - 3810 kV/in	Average value: 34.1 kV/mm Grade Count:15
Dissipation Factor	0.0000200 - 0.00100	0.0000200 - 0.00100	Average value: 0.000228 Grade Count:14

Thermal Properties	Metric	English	Comments
CTE, linear	34.2 - 200 µm/m-°C	19.0 - 111 µin/in-°F	Average value: 116 µm/m-°C Grade Count:11
Melting Point	26.0 - 135 °C	78.8 - 275 °F	Average value: 127 °C Grade Count:40
Maximum Service Temperature, Air	80.0 - 120 °C	176 - 248 °F	Average value: 91.3 °C Grade Count:3
Deflection Temperature at 0.46 MPa (66 psi)	62.0 - 143 °C	144 - 290 °F	Average value: 81.4 °C Grade Count:33
Deflection Temperature at 1.8 MPa (264 psi)	42.0 - 127 °C	108 - 260 °F	Average value: 64.0 °C Grade Count:15
Vicat Softening Point	70.0 - 133 °C	158 - 271 °F	Average value: 116 °C Grade Count:67
Minimum Service Temperature, Air	-100 - -75.0 °C	-148 - -103 °F	Average value: -91.7 °C Grade Count:3
Brittleness Temperature	-118 - 76.0 °C	-180 - 169 °F	Average value: -78.9 °C Grade Count:50
Flammability, UL94	HB - V-0	HB - V-0	Grade Count:4

Processing Properties	Metric	English	Comments
Processing Temperature	180 - 255 °C	356 - 491 °F	Average value: 214 °C Grade Count:7
Adapter Temperature	210 - 260 °C	410 - 500 °F	Average value: 247 °C Grade Count:7
Die Temperature	60.0 - 260 °C	140 - 500 °F	Average value: 235 °C Grade Count:8
Melt Temperature	124 - 321 °C	255 - 610 °F	Average value: 212 °C Grade Count:52
Head Temperature	191 - 235 °C	375 - 455 °F	Average value: 212 °C Grade Count:3
Mold Temperature	15.6 - 65.6 °C	60.0 - 150 °F	Average value: 34.1 °C Grade Count:3
Drying Temperature	76.7 - 79.4 °C	170 - 175 °F	Average value: 79.1 °C Grade Count:4

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.